

Scrum Project Report: Peer Skill Exchange & Mentorship Network

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Problem Statement: #56 — Peer Skill Exchange & Mentorship Network

Course: Software Engineering Lab (UE24CS252AA) — Lab 2: Agile Backlog Creation & Sprint Simulation in Jira

Project Key: SPS56

1. Project Objective & Methodology

This laboratory session models an Agile Scrum software development lifecycle using Atlassian Jira. High-level functional requirements from Problem Statement #56 (Peer Skill Exchange & Mentorship Network) are converted into functional Epics, deconstructed into User Stories using standard Agile conventions ("As a... I want to... So that..."), estimated using Planning Poker with the Fibonacci sequence, and tracked through a time-boxed sprint simulation evaluated via an Agile Burndown Chart.

2. Backlog Creation & Epic Hierarchy

The system scope is categorized into three major functional themes (Epics) to group related end-user capabilities:

- Epic 1: Time-Banking Ledger & Credit Transactions: Manages non-monetary time credits, escrow holding mechanisms during active reservations, and automated ledger settlements.
- Epic 2: Mentor Discovery & Session Scheduling: Encompasses mentor profile queries, multi-criteria filtering, slot booking, and calendar integration.
- Epic 3: Session Validation, Feedback & Disputes: Governs dual-confirmation completion handshakes, rating mechanisms, and dispute resolution workflows.

Seven backlog work items (5 Functional Requirements and 2 Non-Functional Requirements) were mapped under these parent Epics:

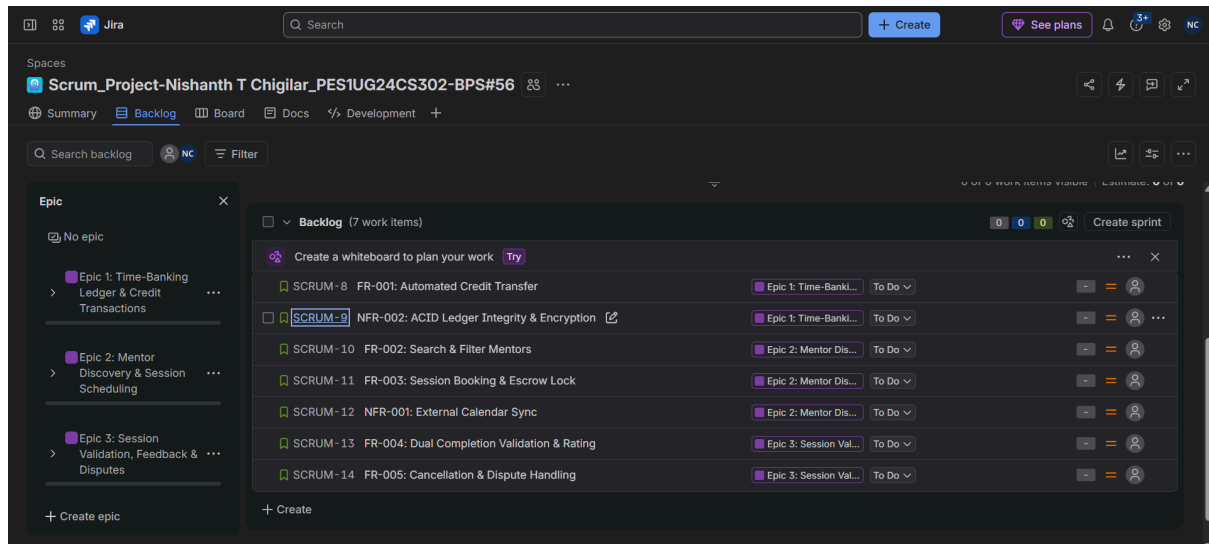


Figure 1: Jira Backlog interface displaying the expanded Epic panel on the left with all seven User Stories created under their respective parent Epics.

- **SCRUM-8 (FR-001): Automated Credit Transfer** — As a Skill Mentor, I want 1 time credit transferred to my balance upon session verification, so that my teaching time is fairly rewarded. (Parent: Epic 1)
- **SCRUM-9 (NFR-002): ACID Ledger Integrity & Encryption** — As a Platform Admin, I want all credit ledger transactions to execute with strict ACID consistency and TLS/AES encryption, so that credit balances remain tamper-proof. (Parent: Epic 1)
- **SCRUM-10 (FR-002): Search & Filter Mentors** — As a Learner Member, I want to search and filter mentor profiles by skill tags, ratings, and open slots, so that I can easily find a suitable mentor. (Parent: Epic 2)
- **SCRUM-11 (FR-003): Session Booking & Escrow Lock** — As a Learner Member, I want to book an available mentor slot by holding 1 credit in escrow, so that my session is guaranteed without immediate deduction. (Parent: Epic 2)
- **SCRUM-12 (NFR-001): External Calendar Sync** — As a Learner Member, I want confirmed sessions to export an .ics feed to Google and Outlook calendars within 5 seconds, so that my external schedule stays up to date. (Parent: Epic 2)
- **SCRUM-13 (FR-004): Dual Completion Validation & Rating** — As a Learner Member, I want to confirm session completion and submit a 1–5 star rating with comments within 48 hours, so that the mentor receives credit and the community sees verified feedback. (Parent: Epic 3)
- **SCRUM-14 (FR-005): Cancellation & Dispute Handling** — As a Community Member, I want to cancel sessions >24 hours prior for a full refund or raise a dispute for a no-show, so that my time credits are protected. (Parent: Epic 3)

3. Backlog Prioritization & Story Point Estimation

Effort estimation was performed using the non-linear Fibonacci sequence (1, 2, 3, 5, 8, 13, ...) through an Agile Planning Poker exercise. Points reflect technical complexity, implementation effort, and operational uncertainty rather than raw developer hours. The product backlog was vertically ranked so that foundational discovery and core transaction items precede secondary integrations.

Issue Key & Type	Work Item Summary	Priority	Story Points	Estimation Rationale & Complexity Factors
SCRUM-10 (Story)	FR-002: Search & Filter Mentors	Medium ▾	3	Read-heavy database indexing and basic client-side filter controls; low architectural risk.
SCRUM-11 (Story)	FR-003: Session Booking & Escrow Lock	Medium ▾	5	Requires balance pre-authorization and conditional state transitions into an escrow holding table.
SCRUM-8 (Story)	FR-001: Automated Credit Transfer	Medium ▾	5	Core time-banking engine; requires debit/credit ledger writes

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				and session state validation.
SCRUM-13 (Story)	FR-004: Dual Completion Validation & Rating	Medium ▾	3	Standard form processing for reviews, star ratings, and dual-acknowledgment triggers.
SCRUM-12 (Story)	NFR-001: External Calendar Sync	Medium ▾	3	Standardized RFC 5545 iCalendar (.ics) generation with basic calendar export handling.
SCRUM-14 (Story)	FR-005: Cancellation & Dispute Handling	Medium ▾	5	Edge-case state engine managing 24-hour cutoff windows, automatic escrow refunding,

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				and ticket logging.
SCRUM-9 (Story)	NFR-002: ACID Ledger Integrity & Encryption	Medium ▾	8	High technical complexity; requires isolated database transactions, row-level locking, and cryptographic hashing.
Total	Total Backlog Scope	— ▾	32	Complete story point baseline across all 7 user stories.

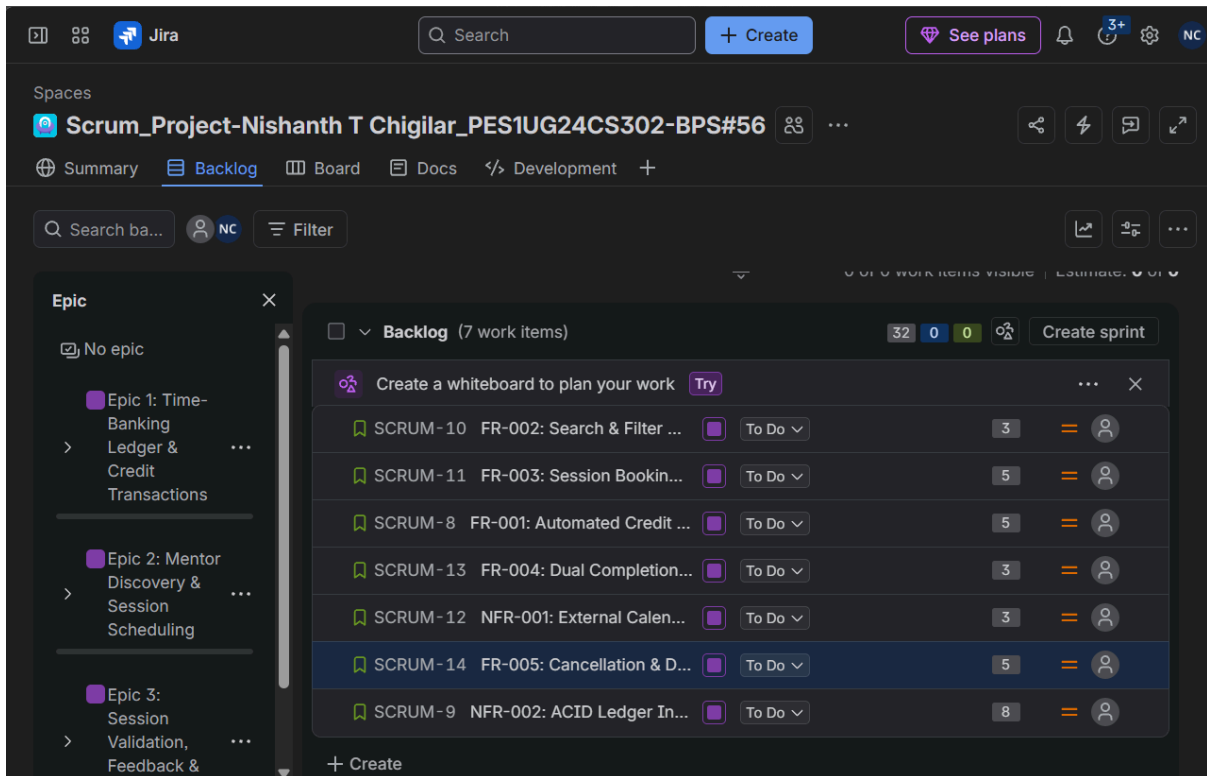


Figure 2: Prioritized Jira Backlog displaying the 32 total estimated points, Fibonacci estimate badges (3, 5, 8), and ranked backlog ordering.

4. Sprint 1 Simulation & Active Board Execution

A mini-sprint (**SCRUM Sprint 1**) was initiated with a planned duration of 1 week and a targeted sprint goal:

- **Sprint Name:** **SCRUM Sprint 1**
- **Sprint Goal:** Implement core mentor discovery, slot booking escrow, and verified credit transfer.
- **Committed Scope:** 4 User Stories (**SCRUM-10**, **SCRUM-11**, **SCRUM-8**, **SCRUM-13**) totaling **16 Story Points**.
- **Workflow Simulation:** Tasks were progressed systematically across **To Do** → **In Progress** → **Done** columns to simulate active development and verification.

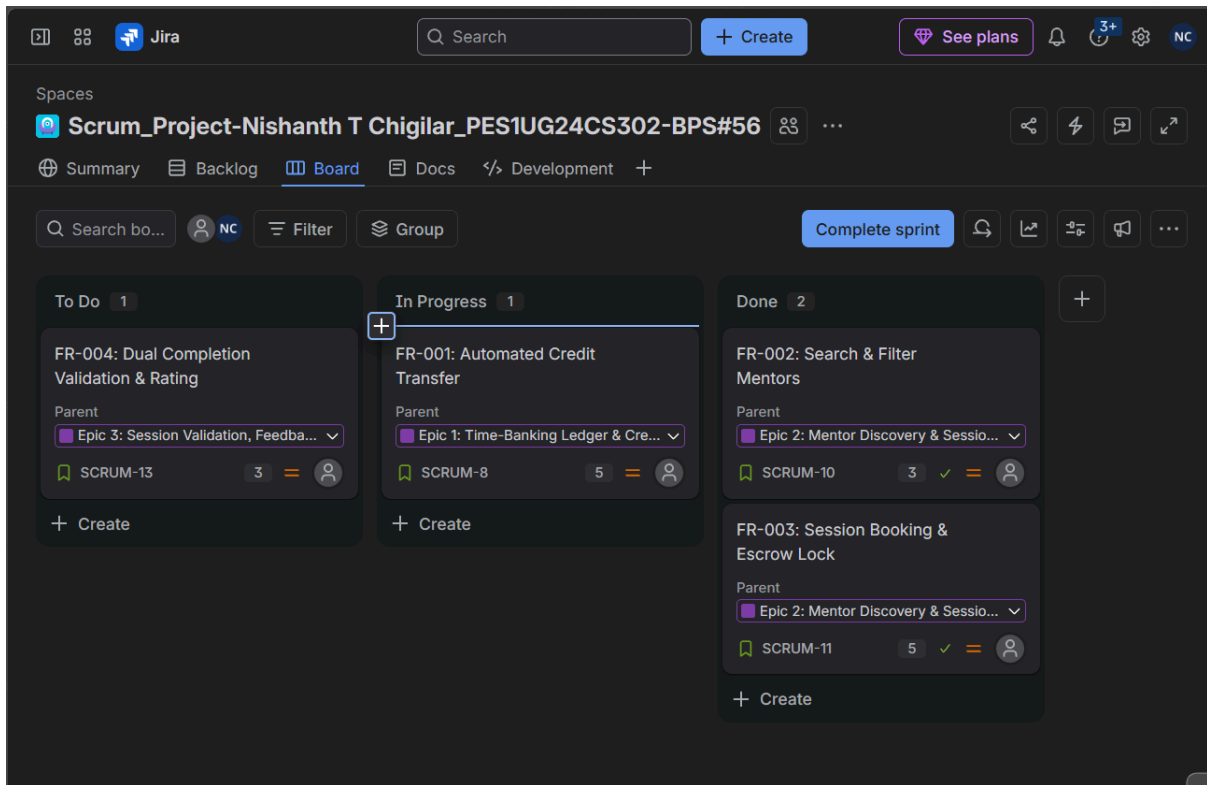


Figure 3: Jira Active Sprint Board showing real-time work-in-progress distribution across To Do (SCRUM-13), In Progress (SCRUM-8), and Done (SCRUM-10, SCRUM-11).

5. Burndown Chart Analysis

After all in-scope sprint items were moved to the Done column, the sprint was formally closed via the Complete sprint action. Progress was monitored through the Jira Sprint Burndown Chart configured with Story points as the estimation statistic.

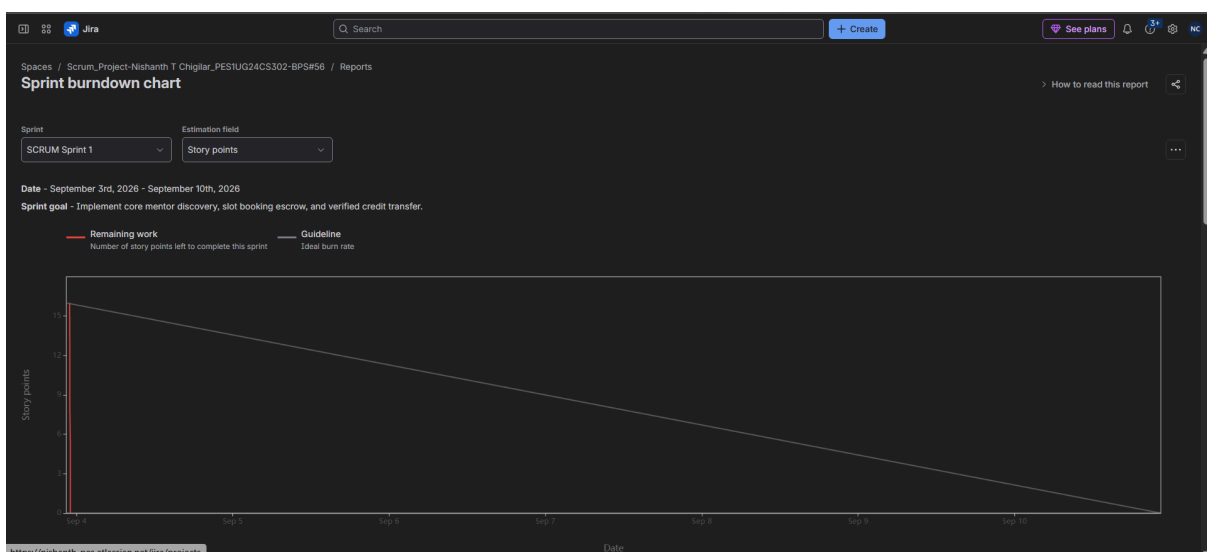


Figure 4: Sprint Burndown Chart illustrating the burn rate of the 16 committed story

points dropping to zero against the ideal guideline. The horizontal X-axis plots the sprint timeline (September 3, 2026 – September 10, 2026), while the vertical Y-axis tracks remaining estimated story points. The grey diagonal line represents the theoretical linear burn rate required to complete work evenly over the sprint. The red line captures actual remaining points, showing the completion of all 16 story points.

6. Agile Reflection Questions

1. Did your estimations reflect the actual effort?

Yes, the relative point assignments aligned well with the technical depth of each requirement. Applying the non-linear Fibonacci sequence through Planning Poker enabled a clear distinction between routine UI/filtering features and complex state-driven features. A task such as mentor search (SCRUM-10, 3 points) only required query parameter binding and pagination, whereas transactional escrow locking (SCRUM-11, 5 points) and credit transfer (SCRUM-8, 5 points) demanded rigorous validation of balances, concurrency safeguards, and multi-table atomic operations.

2. Was your backlog well-prioritized?

Yes, the backlog prioritization reflected a logical dependency flow and prioritized business value. Discovery (FR-002) and booking with escrow (FR-003) were ranked at the top because a peer exchange platform cannot operate without initial session scheduling. Credit settlement (FR-001) and completion sign-off (FR-004) directly followed to complete the core user transaction. Peripheral integrations (external calendar feeds) and edge-case exceptions (cancellations and disputes) were intentionally prioritized lower for subsequent sprint cycles.

3. How did your simulated sprint align with your plan?

The sprint simulation tracked the defined sprint goal cleanly. By constraining Sprint 1 to 16 story points—exactly half of the overall backlog capacity—the team focused strictly on delivering a functional core transaction path. Work items transitioned sequentially through To Do, In Progress, and Done without introducing unplanned user stories or scope creep, resulting in a 100% completion rate for committed deliverables.

4. What insights did the burndown chart give about your team's capacity?

The burndown chart confirmed that a commitment of 16 story points was a manageable velocity for this sprint scope. However, because tasks were moved to

Done in batches during the simulation, the actual burn line exhibited abrupt step drops rather than an even downward progression along the linear guideline. In an operational environment, this pattern highlights the need to deconstruct 5-point and 8-point stories into smaller 1–2 point daily subtasks, ensuring continuous incremental delivery and more accurate real-time capacity tracking.